

Plastic Shredder and Injection Prototype

Sidharth S, Manu Nair
Department of Mechanical Engineering
Sree Chitra Thirunal College of Engineering

Abstract— Plastic causes serious environmental problems. Although they are not intrinsically dangerous, they take up a huge amount of space in landfills and they are made from a non-renewable resources, namely fossil fuels. For this reason it is important that, where possible plastics are reused or recycled. The use of plastic is increased now days in many industries like automobile, packaging, medical, etc. The reason behind this is that the plastic made things are quite easier to manufacture, handle and reliable to use. So the plastic goods manufacturing industries are striving hard to produce good quality products at large scale and cheaper cost. The hydraulically operated machines solve the problem, but they are too costlier for small scale and medium scale industries. This project deals with fabrication of plastic shredder and injection prototype. It includes assembly of mechanical components like hopper, frame, heating-coil, power supply etc. The Plastic waste is first converted into plastic flakes and these flakes can either be disposed of efficiently or are used in the injection machine to produce different types of products.

I. INTRODUCTION

Plastics are everywhere. A plastic is any material that can be shaped or molded into any form -- some are naturally occurring, but most are man-made. Plastics are made from oil. Oil is a carbon-rich raw material, and plastics are large carbon-containing compounds. They're large molecules called polymers, which are composed of repeating units of shorter carbon-containing compounds called monomers. Chemists combine various types of monomers in many different arrangements to make an almost infinite variety of plastics with different chemical properties. Most plastic is chemically inert and will not react chemically with other substances -- you can store alcohol, soap, water, acid or gasoline in a plastic container without dissolving the container itself. Plastic can be molded into an almost infinite variety of shapes, so you can find it in toys, cups, bottles, utensils, wiring, cars, even in bubble gum. Plastics have revolutionized the world. Due to their relatively low cost, ease of manufacture, versatility, and imperviousness to water, plastics are used in an enormous and expanding range of products, from paper clips to spaceships. They have already displaced many traditional materials, such as wood, stone, horn and bone, leather, paper, metal, glass, and ceramic, in most of their former uses. In developed countries, about a third of plastic is used in packaging and another third in buildings such as piping used in plumbing etc.

II. TYPES OF PLASTIC

- 1 - Polyethylene terephthalate (PET)
 - 2 - High-density polyethylene (HDPE)
 - 3 - Polyvinyl chloride (PVC)
 - 4 - Low-density polyethylene (LDPE)
 - 5 - Polypropylene (PP)
 - 6 - Polystyrene (PS)
 - 7 - Other resins, like acrylonitrile butadiene styrene (ABS)
- PET is most commonly seen in plastic soda bottles and Mylar film.
- HDPE is what milk jugs are made of, and is also used in things like plastic gas cans.
- PVC is the base for any form of vinyl (from siding to seat covers), as well as many soft toys.
- LDPE is found in things like plastic food containers.
- The difference between HDPE and LDPE is that LDPE is softer, more flexible and melts at a lower temperature than HDPE.
- PP is a lightweight plastic and has a relatively high melting point -- it is used in things like luggage and plastic trim in cars, as well as food containers.
- PS is commonly known by the Styrofoam brand name and is used in everything from coffee cups to coolers.

III. RECYCLING PLASTICS

Oil-based plastics don't degrade, but many types (including PP, LDPE, HDPE, PET, and PVC) can be recycled. Each type has a code and identifying number, but some plastics aren't as economically feasible to recycle.

- Once collected, plastics go through the following steps
- Inspection to weed out contaminants and inappropriate types of plastic
- Shredding and washing
- Separation based on density
- Drying
- Melting
- Draining through fine screens to remove more contaminants
- Cooling and shredding into pellets
- Selling back to plastic companies

The discovery of plastics revolutionized our society by introducing a huge variety of lightweight, strong, flexible

products with many uses. Although plastics do pose disposal problems, recycling is always a possibility. Furthermore, new research into biopolymers may produce new bioplastic products from renewable resources that are biodegradable and easier on our environment.

Type	name	properties	common uses	burning
PET 	polyethylene terephthalate	clear, tough, solvent resistant, barrier to gas and moisture, softens at 80°	Soft drink, water bottles, salad domes, biscuit trays, food containers	yellow flame little smoke
HDPE 	high-density polyethylene	Hard to semi-flexible, resistant to chemicals and moisture, waxy surface, softens at 75°	Shopping bags, freezer bags, milk bottles, juice bottles, icecream containers, shampoo, crates	difficult to ignite smells like candle
PVC 	polyvinyl chloride	Strong, tough, can be clear and solvent softens at 80°	Cosmetic containers, electrical conduit, plumbing pipes, blister packs, roof sheathing, garden hose	yellow flame green spurts
LDPE 	low-density polyethylene	Soft, flexible, waxy surface, scratches easily, softens at 70°	Cling wrap, garbage bags, squeeze bottles, refuse bags, mulch film	difficult to ignite smells like candle
PP 	polypropylene	Hard but still flexible, waxy surface, translucent, withstands solvents, softens at 140°	Bottles, icecream tubes, straws, flowerpots, dishes, garden furniture, food containers	blue yellow tipped flame
PS 	polystyrene	Clear, glassy, opaque, semi tough, softens at 90°	CD cases, plastic cutlery, imitation glass, foamed meat trays, brittle toys	dense smoke
OTHER 	all other plastics	Properties depend on the type of plastic	automotive, electronics, packaging	all other plastics

The problem with recycling of plastics is as different types of plastics melt at different temperatures, the recycling process is time consuming and financially not viable for small scale recycling.

Plastic Abbreviation - Brand name	Thermal Properties				Strength		Density g/cc
	Tm °C	Tg °C	Td °C	Cte ppm/°C	Tensile psi	Compressive psi	
PET - Polyethyleneterephthalate	245 265	73 80	21 38	65	7000 10500	11000 15000	1.29 1.40
LDPE - Low density polyethylene	98 115	-25	40 44	100 220	1200 4550		0.917 0.932
HDPE - High density polyethylene	130 137		79 91	59 110	3200 4500	2700 3600	0.952 0.965
PP - polypropylene	168 175	-20	107 121	81 100	4500 6000	5500 8000	0.900 0.910
PVC - polyvinylchloride		75 105	57 82	50 100	5900 7500	8000 13000	1.30 1.58
PS - polystyrene		74 105	68 96	50 83	5200 7500	12000 13000	1.04 1.05

Tm - crystalline melting temperature (some plastics have no crystallinity and are said to be amorphous).

Tg - glass transition temperature (the plastic becomes brittle below this temperature).

Td - heat distortion temperature under a 66 psi load.

Cte - coefficient of linear thermal expansion.

Tensile Strength - load necessary to pull a sample of the plastic apart. Compressive Strength - load necessary to crush a sample of the plastic. Density - aka specific gravity mass of plastic per unit volume.

Plastic materials can be recycled in a variety of ways and the ease of recycling varies among polymer type, package design and product type. For example, rigid containers consisting of a single polymer are simpler and more economic to recycle than multi-layer and multi-component packages.

A major challenge for producing recycled resins from plastic wastes is that most different plastic types are not compatible with each other because of inherent immiscibility at the molecular level, and differences in processing requirements at a macro-scale. For example, a small amount of PVC contaminant present in a PET recycle stream will degrade the recycled PET resin owing to evolution of hydrochloric acid gas from the PVC at a higher temperature required to melt and reprocess PET. Conversely, PET in a PVC recycle stream will form solid lumps of undispersed crystalline PET, which significantly reduces the value of the recycled material.

Generally, In large scale industries preliminary processes are carried out to increase the efficiency of the recycling process, They are:

(a) Collection Of Plastics

Collection of plastic wastes can be done by 'bring-schemes'. Bring-schemes tend to result in low collection rates in the absence of either highly committed public behaviour or deposit-refund schemes that impose a direct economic incentive to participate. While kerbside collection schemes have been very successful at recovering plastic bottle packaging from homes, in terms of the overall consumption typically only 30–40% of post-consumer plastic bottles are recovered, as a lot of this sort of packaging comes from food and beverage consumed away from home. For this reason, it is important to develop effective 'on-the-go' and 'office recycling' collection schemes if overall collection rates for plastic packaging are to increase.

(b) Sorting

Sorting of co-mingled rigid recyclables occurs by both automatic and manual methods. Automated pre-sorting is usually sufficient to result in a plastics stream separate from glass, metals and paper (other than when attached, e.g. as labels and closures). Generally, clear PET and unpigmented HDPE milk bottles are positively identified and separated out of the stream. Automatic sorting of containers is now widely used by material recovery facility operators and also by many plastic recycling facilities. These systems generally use Fourier-transform near-infrared (FT-NIR) spectroscopy for polymer type analysis and also use optical colour recognition camera systems to sort the streams into clear and coloured fractions. Optical sorters can be used to differentiate between clear, light blue, dark blue, green and other coloured PET containers. Sorting performance can be maximized using multiple detectors, and sorting in series. Other sorting technologies include X-ray detection, which is used for separation of PVC

containers, which are 59 per cent chlorine by weight and so can be easily distinguished.

(c) Size reduction and cleaning

Rigid plastics are typically ground into flakes and cleaned to remove food residues, pulp fibres and adhesives. The latest generation of wash plants use only 2–3 m³ of water per tonne of material, about one-half of that of previous equipment. Innovative technologies for the removal of organics and surface contaminants from flakes include ‘dry-cleaning’, which cleans surfaces through friction without using water

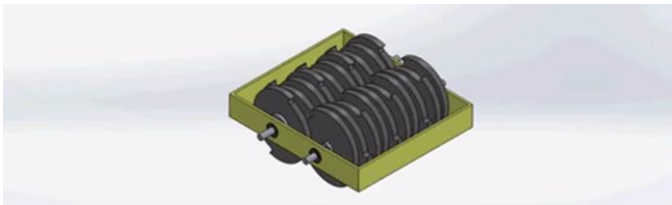
IV .WORKINING OF PROTOTYPE

Before we can Recycle the plastics it is necessary to collect them and sort them in a systematic way so as to reduce any hassle in the processes. This sorting and arranging is done based on the type of plastics and their properties like melting temperature etc.

Our Prototype consists mainly of 2 different machines that utilize different processes for making the recycling efficient and easy. The different machines are:

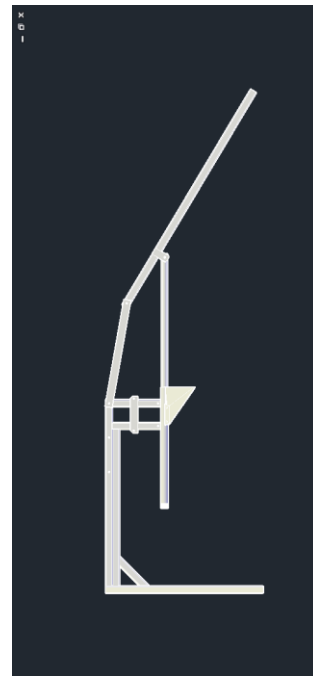
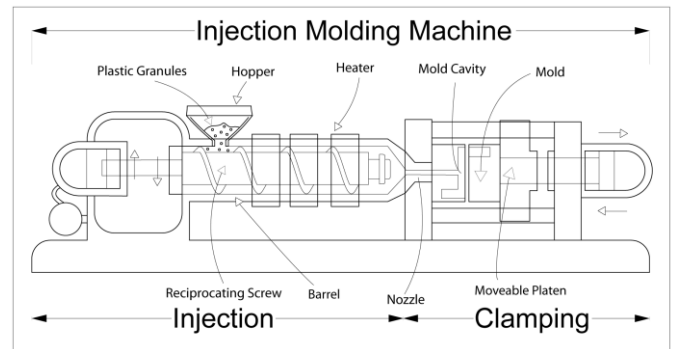
1. Plastic Shredder

This machine chops the plastic waste into small plastic flakes and these flakes will be used in the machine to create new plastic products. The size of plastics that can be shredded depends on the size of shredder and the power supply.



2. Injection Machine

The injection machine injection machine injects the heated plastic into a mould. The shredded plastic is given to the machine through a hopper. Electronics are used to control and set the temperatures required for injection.



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